

Measurement - Introduction

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MEASUREMENT helps you determine how big or how small things in the real world are. In this unit, you will learn how to measure length, weight, capacity, time, and temperature.

All measurements must have **units**. Without units, measurements are just numbers that don't have much meaning. You can add, subtract, multiply, and divide measurements as long as you pay attention to the units.

In this unit, you'll learn how to use both customary and metric units of measure. You'll also learn important tools for working with measurements in word problems.

USING MEASUREMENT

You use measurement when you:

- ✓ • Use a ruler, yardstick, or tape measure
- ✓ • Measure quantities in a recipe
- ✓ • Stand on a scale
- ✓ • Use a stopwatch to time a lap
- ✓ • Estimate how long a trip will take

MATH TALK

we'll look at these in lesson #2.

Meters, liters, and grams are the basic units of the metric system.

Basic Units

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In this lesson, we will cover the following:

- Length ✓
- Capacity ✓
- Weight ✓
- Metric *next lesson*
- Conversions
- Time & Temperature ✓
- Distance, Rate, Time ✓

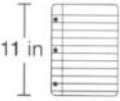
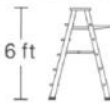
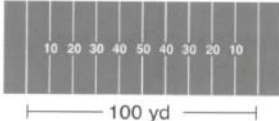
Basic Units of Length

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We must learn the American system of measurement because the GED uses that system (not because it's better than other systems!).

CUSTOMARY UNITS OF LENGTH

The measuring system most often used in the United States is called the **U.S. customary system**. The most often used customary units of length are shown below.

Unit	Sample Use	Comparing Units
inch (in)		$1 \text{ in} = \frac{1}{12} \text{ ft}$
foot (ft)		$1 \text{ ft} = 12 \text{ in}$ $1 \text{ ft} = \frac{1}{3} \text{ yd}$
yard (yd)		$1 \text{ yd} = 3 \text{ ft}$ $1 \text{ yd} = 36 \text{ in}$
mile (mi)		$1 \text{ mi} = 1,760 \text{ yd}$ $1 \text{ mi} = 5,280 \text{ ft}$

your "pinkie"
finger is
about 2 inches

about 1.6 km

Length, cont'd...

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Sometimes measurements need to be written in larger or smaller units. Use your common sense to choose an operation. Remember, multiplication generally makes more things. Division makes fewer things.

To change

Smaller to
Larger Units $\div$

Larger to
Smaller Units $\times$

CONVERTING TO LARGER UNITS

Example Maria wonders if the couch she wants to buy will fit against a wall that is 7 ft long. The couch is 76 in long. Convert its length to feet and inches.

Step 1

Set up the problem.

76 in = ____ ft ____ in

Step 2

Divide by the number of
smaller units.
(12 in = 1 ft)

$76 \text{ in} \div 12 = 6 \text{ R}4$

Step 3

Write the whole number
as the larger unit and the
remainder as the smaller unit.

76 in = **6 ft 4 in**

6 ft 4 in is less than 7 ft. The couch will fit against the wall.

wall

$$\begin{aligned} 7 \text{ ft} &= 7(12) \text{ in} \\ &= 84 \text{ in} \\ \text{yes!!} \end{aligned}$$

couch (sofa)

$$\begin{aligned} 76 \text{ in} &= 76 / 12 \\ &= 6.33 \text{ ft} \\ &= 6 \text{ ft and } (.33)(12 \text{ in}) \\ &= 6 \text{ ft, } 4 \text{ in} \\ \text{yes!!} \end{aligned}$$

Both are suitable
solutions.

CONVERTING TO SMALLER UNITS

Example Roberta needed $6\frac{1}{4}$ feet of material for a window covering. How many inches of material does she need?

Step 1

Set up the problem.

$$6\frac{1}{4} \text{ ft} = \underline{\hspace{1cm}} \text{ in}$$

Roberta will need **75 inches** of material.

Step 2

Multiply by the number of smaller units.

$$6\frac{1}{4} \times 12 = \frac{25}{4} \times \frac{12^3}{1} = 75 \text{ in}$$

$$6.25 \text{ ft} = 6(12'') + (.25)(12'')$$

$$= 72'' + 3''$$

$$= 75 \text{ in}$$

Both are printable solutions.

$$6.25 \text{ ft} = 6.25(12'')$$

$$= 75 \text{ in}$$

Length - Examples Set #1

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Change each length to the larger unit indicated.

1. 28 in = 2 ft 4 in
(Hint: 1 foot = ? inches)

$$28 \div 12 = 2 \frac{4}{12}$$

47 in = 3 ft 11 in

$$47 \div 12 = 3 \frac{11}{12}$$

16 ft = 5 yd 1 ft
3 ft = 1 yd

$$16 \div 3 = 5 \frac{1}{3}$$

2. 23 ft = 7 yd 2 ft

$$23 \div 3 = 7 \frac{2}{3}$$

81 in = 2 yd 9 in

36 in = 1 yd

$$81 \div 36 = 2 \frac{9}{36}$$

110 in = 3 yd 2 in

$$110 \div 36 = 3 \frac{2}{36}$$

scratch

$$\frac{36}{36} = 1$$

Change each length to the smaller unit indicated.

3. 3 ft = 36 in

$$3 \times 12 = 36$$

5 $\frac{1}{6}$ ft = 62 in

This is NOT $5 \times \frac{1}{6}$.

This is $5 + \frac{1}{6}$.

$$(5 + \frac{1}{6})(12) =$$

$$5(12) + \frac{1}{6}(12) =$$

$$60 + 2 = 62$$

1 $\frac{1}{3}$ yd = 4 ft

same here.

$$(1 + \frac{1}{3})(3) =$$

$$(1)(3) + (\frac{1}{3})(3) =$$

$$3 + 1 = 4$$

4. $4 \frac{3}{4}$ yd = 171 in

$$4 \text{ yd} + \frac{3}{4} \text{ yd}$$

$$= (4)(36 \text{ in})$$

$$+ (\frac{3}{4})(36 \text{ in})$$

$$= 144 + 27$$

$$= 171$$

$1 \frac{1}{5}$ mi = 2112 yd

1 mi = 5280 ft

= 1760 yd

$$1 \text{ mi} + \frac{1}{5} \text{ mi}$$

$$= 1(1760 \text{ yd})$$

$$+ \frac{1}{5}(1760 \text{ yd})$$

$$= 1760 + 352$$

$$= 2112$$

$2 \frac{5}{6}$ mi = 14960 ft

2 mi + $\frac{5}{6}$ mi

$$= 2(5280) + \frac{5}{6}(5280)$$

$$= 10560 + 4400$$

$$= 14960$$

Length - Examples Set #2

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Solve the following word problems.

5. George needed 129 feet of fencing to protect his pool. The hardware store sells the fencing by the yard. How many yards of fencing is 129 feet?

$$\begin{aligned} 3 \text{ ft} &= 1 \text{ yd} \\ 129 \text{ ft} &= (129/3) \text{ yd} \\ &= 43 \text{ yd} \end{aligned}$$

6. Lila must mail several boxes of personnel forms to the branch office. She estimates she needs 233 inches of packing tape. If the store sells tape by the whole yard only, how many yards of tape does Lila need?

$$\begin{aligned} 36 \text{ in} &= 1 \text{ yd} \\ 233 \text{ in} &= (233/36) \text{ yd} \\ &= 6.47\bar{2} \\ &= 7 \text{ yds} \end{aligned}$$

7. Marcus is the tallest person in his school at $6\frac{1}{2}$ feet tall. How many inches tall is Marcus?

$$\begin{aligned} 1 \text{ ft} &= 12 \text{ in} \\ 6\frac{1}{2} \text{ ft} &= 6.5 (12) \text{ in} \\ &= 78 \text{ in} \end{aligned}$$

8. **Multiple Answers** Rashaad passed for 12,534 yards during his football career. List his passing yards in as many different units of length as you can (in, ft, mi).

$$\begin{aligned} 1 \text{ yd} &= 36 \text{ in} \\ 12534 \text{ yds} &= (12534)(36) \text{ in} \\ &= 451,224 \text{ in} \end{aligned}$$

$$\begin{aligned} 1 \text{ yd} &= 3 \text{ ft} \\ 12534 \text{ yds} &= (12534)(3) \text{ ft} \\ &= 37,602 \text{ ft} \end{aligned}$$

$$\begin{aligned} 1760 \text{ yd} &= 1 \text{ mi} \\ 12534 \text{ yd} &= 12534 / 1760 \text{ mi} \\ &= 7.12 \text{ mi} \end{aligned}$$

WORKING WITH LENGTH

When working with length, add or subtract each unit separately. With addition and subtraction, estimation helps you know what to expect. If you quickly add or subtract the larger units first, you can tell if your answer makes sense. However, estimation isn't always useful when you multiply or divide.

ADDING UNITS OF LENGTH

Example The manager of Go-Mart wants a display measuring 4 ft 5 in wide placed next to a display measuring 2 ft 8 in wide. How much space will the two displays take up?



Step 1
Set up the problem.

$$\begin{array}{r} 4 \text{ ft } 5 \text{ in} \\ + 2 \text{ ft } 8 \text{ in} \\ \hline \end{array}$$

Step 2
Estimate first by adding the larger units.

$$4 \text{ ft} + 2 \text{ ft} = 6 \text{ ft}$$

The answer should be slightly more than 6 ft.

Step 3
Add the units separately.

$$\begin{array}{r} 4 \text{ ft } 5 \text{ in} \\ + 2 \text{ ft } 8 \text{ in} \\ \hline 6 \text{ ft } 13 \text{ in} \end{array}$$

$$\begin{array}{r} 4' 5'' \\ + 2' 8'' \\ \hline 6' 13'' \\ = 7' 1'' \end{array}$$

7 ft 1 in is slightly more than 6 ft. The two displays will take up **7 ft 1 in**.

SUBTRACTING UNITS OF LENGTH

Example The owners of Go-Mart want to install a 12-ft-long cooler. The space they have in mind is 10 ft 8 in long. How much more space is needed for the cooler to fit?

Step 1
Set up the problem.

$$\begin{array}{r} 12 \text{ ft} \\ - 10 \text{ ft } 8 \text{ in} \\ \hline \end{array}$$

Step 2
Estimate first by subtracting the larger units.

$$12 \text{ ft} - 10 \text{ ft} = 2 \text{ ft}$$

The answer should be slightly less than 2 ft.

Step 3
Regroup from the larger unit, if necessary.

$$12 \text{ ft} = 11 \text{ ft } 12 \text{ in}$$

$$\begin{array}{r} 12 \text{ ft} \\ - 10 \text{ ft } 8 \text{ in} \\ \hline 11 \text{ ft } 12 \text{ in} \\ - 10 \text{ ft } 8 \text{ in} \\ \hline 1 \text{ ft } 4 \text{ in} \end{array}$$

1 ft 4 in is slightly less than 2 ft. Go-Mart needs **1 ft 4 in** more space.

Length, cont'd...

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MULTIPLYING UNITS OF LENGTH

Example Anna needs a wooden dowel that she can cut into 5 pieces. Each piece must be 1 ft 3 in. How long should the wooden dowel be?

Step 1

Set up the problem.

$$\begin{array}{r} 1 \text{ ft } 3 \text{ in} \\ \times \quad 5 \\ \hline \end{array}$$

Step 2

Multiply the number of each unit separately.

$$\begin{array}{r} 1 \text{ ft } 3 \text{ in} \\ \times \quad 5 \\ \hline 5 \text{ ft } 15 \text{ in} \end{array}$$

Step 3

Simplify, if possible.

$$\begin{array}{l} 5 \text{ ft } 15 \text{ in} = 5 \text{ ft} + 1 \text{ ft} + 3 \text{ in} = \\ 6 \text{ ft } 3 \text{ in} \end{array}$$

The wooden dowel should be **6 ft 3 in**.

5 pieces each 1 ft 3 in
each 12 + 3 in
each 15 in

$$\begin{array}{l} \text{total length} \\ = 5(15) \\ = 75 \text{ in} \\ = 72 \text{ in} + 3 \text{ in} \\ = 6 \text{ ft } 3 \text{ in} \end{array}$$

DIVIDING UNITS OF LENGTH

Example If you divide a 13-ft-3-in board into 3 equal pieces, how long is each piece?

Step 1

Set up the problem.

$$3 \overline{) 13 \text{ ft } 3 \text{ in}}$$

Step 2

Divide the larger unit.

$$\begin{array}{r} 4 \\ 3 \overline{) 13 \text{ ft}} \\ - 12 \\ \hline 1 \end{array}$$

Remainder:
1 ft = 12 in

Step 3

Regroup and add any remainder to the smaller unit. Divide.

$$\begin{array}{r} 12 \text{ in} + 3 \text{ in} = 15 \text{ in} \\ 5 \\ 3 \overline{) 15 \text{ in}} \\ - 15 \\ \hline 0 \end{array}$$

Step 4

Combine the amounts from Steps 2 and 3 to get the answer.

$$\begin{array}{l} 4 \text{ ft} + 5 \text{ in} = \\ 4 \text{ ft } 5 \text{ in} \end{array}$$

Each piece should be **4 ft 5 in** long.

$$\begin{aligned} &13 \text{ ft } 3 \text{ in} \\ &= 13(12) + 3 \\ &= 156 + 3 \\ &= 159 \text{ in} \\ &159 \text{ in} \div 3 \\ &= 53 \text{ in} \\ &= 48 + 5 \text{ in} \\ &= 4 \text{ ft } 5 \text{ in} \end{aligned}$$

Length - Examples Set #3

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Add or subtract the following lengths. Regroup when necessary.

1.
$$\begin{array}{r} 6 \text{ ft } 4 \text{ in} \\ + 3 \text{ ft } 5 \text{ in} \\ \hline \end{array}$$

$9 \text{ ft } 9 \text{ in}$

$$\begin{array}{r} 2 \text{ ft } 10 \text{ in} \\ + 5 \text{ ft } 6 \text{ in} \\ \hline \end{array}$$

$7 \text{ ft } 16 \text{ in}$
 $= 8 \text{ ft } 4 \text{ in}$

$$\begin{array}{r} 7 \text{ ft } 10 \text{ in} \\ + 6 \text{ ft } 11 \text{ in} \\ \hline \end{array}$$

$13 \text{ ft } 21 \text{ in}$
 $= 14 \text{ ft } 9 \text{ in}$

$$\begin{array}{r} 4 \text{ yd } 2 \text{ ft} \\ + 2 \text{ yd } 2 \text{ ft} \\ \hline \end{array}$$

$6 \text{ yd } 4 \text{ ft}$
 $= 7 \text{ yd } 1 \text{ ft}$

2.
$$\begin{array}{r} 4 \text{ ft } 7 \text{ in} \\ - 1 \text{ ft } 3 \text{ in} \\ \hline \end{array}$$

$3 \text{ ft } 4 \text{ in}$

$$\begin{array}{r} 5 \text{ ft } 6 \text{ in} \\ - 2 \text{ ft } 8 \text{ in} \\ \hline \end{array}$$

$4 \text{ ft } 18 \text{ in}$
 $- 2 \text{ ft } 8 \text{ in}$
 $\hline$
 $2 \text{ ft } 10 \text{ in}$

$$\begin{array}{r} 9 \text{ ft } 1 \text{ in} \\ - 4 \text{ ft } 10 \text{ in} \\ \hline \end{array}$$

$8 \text{ ft } 13 \text{ in}$
 $- 4 \text{ ft } 10 \text{ in}$
 $\hline$
 $4 \text{ ft } 3 \text{ in}$

$$\begin{array}{r} 6 \text{ yd } 1 \text{ ft} \\ - 1 \text{ yd } 2 \text{ ft} \\ \hline \end{array}$$

$5 \text{ yd } 4 \text{ ft}$
 $- 1 \text{ yd } 2 \text{ ft}$
 $\hline$
 $4 \text{ yd } 2 \text{ ft}$

Multiply or divide the following lengths.

3.
$$\begin{array}{r} 2 \text{ ft } 4 \text{ in} \\ \times 3 \\ \hline \end{array}$$

$6 \text{ ft } 12 \text{ in}$
 $= 7 \text{ ft}$

$$\begin{array}{r} 7 \text{ ft } 10 \text{ in} \\ \times 6 \\ \hline \end{array}$$

$42 \text{ ft } 60 \text{ in}$
 $= 47 \text{ ft}$

5 ft

4.
$$2 \overline{) 4 \text{ ft } 8 \text{ in}}$$

$= 2 \text{ ft } 4 \text{ in}$

18 ft
 $4 \overline{) 6 \text{ yd } 2 \text{ ft}}$

5 ft
 $4 \overline{) 20 \text{ ft}}$
 $= 1 \text{ yd } 2 \text{ ft}$

Solve the following word problems.

5. A.J. plants 4 bushes in a 6-ft-9-in-long flower bed. Two of the bushes are at opposite ends. If the bushes are equally spaced, how far is each from the next?



$6 \text{ ft } 9 \text{ in} = 81 \text{ in}$

$81 \text{ in} \div 3 = 27 \text{ in}$

$= 2 \text{ ft } 3 \text{ in}$

7. Tanya bought 3 computer cables. Each cable was 6 yd 1 ft. What was the total length of the cables? $\rightarrow 18 \text{ ft}$

$19 \text{ ft}(3) = 57 \text{ ft}$

$= 19 \text{ yds}$

OR, $3 (6 \text{ yd} + 1 \text{ ft})$

$= 18 \text{ yds} + 3 \text{ ft}$

$= 19 \text{ yds}$

6. A junior basketball team has 3 guards. Their heights are 4 ft 2 in, 5 ft 4 in, and 5 ft 9 in. What is the average height of the guards? (Hint: Add and divide.)

$$4'2'' = 50''$$

$$5'4'' = 64''$$

$$5'9'' = 69''$$

$$\text{ave} = \frac{(50 + 64 + 69)}{3}$$

$$= 61''$$

$$= 5'1''$$

8. **Discover** Find 3 whole numbers that divide into 3 ft 8 in and give an answer that does not include fractions of an inch. Then find 3 whole numbers by which to multiply 3 ft 8 in to get a whole number of feet.

$$a) 3\text{ft } 8\text{in} = 44\text{in}$$

$$\{2, 4, 11\}$$

$$b) \text{possible inches:}$$

$$48, 60, 72, 84, 96, 108$$

$$3\text{ft } 8\text{in}$$

$$8 \rightarrow \overset{\times 2}{16}, \overset{\times 3}{24}, 32, 40, \overset{\times 6}{48}, 5$$

$$\therefore 3\text{ft } 8\text{in} \times 3 = 9\text{ft } 24''$$

$$= 11\text{ft}$$

$$3\text{ft } 8\text{in} \times 6 = 18\text{ft } 48''$$

$$= 22\text{ft}$$

$$3\text{ft } 8\text{in} \times 9 = 27\text{ft } 72''$$

$$= 33\text{ft}$$

$$\{3, 6, 9\}$$